

SOUTHERN CALIFORNIA, CA, USA

WMO: 723825

Lat: 34.583N

Lon: 117.383W

Elev: 879

StdP: 91.20

Time Zone: -8.00 (NAP)

Period: 86-17

WBAN: 23131

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -3.6	(c) -2.1	(d) -18.1	(e) 0.8	(f) 6.8	(g) -16.7	(h) 1.0	(i) 6.4	(j) 12.5	(k) 8.4	(l) 11.2	(m) 9.8	(n) 1.1	(o) 170	(p) 0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.5	(c) 38.9	(d) 17.9	(e) 37.7	(f) 17.3	(g) 36.4	(h) 16.9	(i) 20.1	(j) 31.8	(k) 19.3	(l) 32.2	(m) 18.6	(n) 32.5	(o) 4.3	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	13.6	23.9	15.0	11.9	24.4	13.1	10.4	24.9	61.3	30.7	58.6	32.0	56.1	32.5	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.3	(b) 9.5	(c) 8.3	(e) -6.6	(f) 41.7	(g) 2.6	(h) 1.3	(i) -8.5	(j) 42.7	(k) -10.0	(l) 43.4	(m) -11.5	(n) 44.1	(o) -13.4	(p) 45.1		
(a) 11.3	(b) 9.5	(c) 8.3	(e) -8.6	(f) 21.9	(g) 2.3	(h) 1.7	(i) -10.3	(j) 23.1	(k) -11.6	(l) 24.1	(m) -12.9	(n) 25.1	(o) -14.6	(p) 26.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	7.6	10.2	12.9	16.5	19.6	25.1	27.9	27.5	23.9	18.7	12.0	7.1
	DBStd	8.09	3.20	4.21	3.74	3.71	4.10	3.71	2.87	2.32	3.81	4.16	3.91	3.62
	HDD10.0	276	86	45	15	2	0	0	0	0	0	1	24	102
	HDD18.3	1442	333	227	171	77	36	3	0	0	8	46	193	347
	CDD10.0	3000	11	52	105	198	299	453	554	542	418	272	83	13
	CDD18.3	1125	0	1	3	22	76	207	296	284	176	58	2	0
	CDH23.3	15050	0	20	102	395	1035	2772	3996	3721	2158	796	56	1
	CDH26.7	8331	0	2	15	122	442	1612	2441	2231	1141	319	7	0
(14) Wind	WSAvg	2.9	2.2	2.8	3.4	3.6	3.9	3.5	3.0	2.7	2.5	2.3	2.2	2.2
(15) Precipitation	PrecAvg	134	24	38	20	6	5	2	4	5	5	11	8	17
	PrecMax	276	139	142	77	27	41	20	26	28	39	96	29	83
	PrecMin	26	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	78	39	41	26	9	10	5	6	8	10	23	10	24
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.9	26.2	28.6	32.2	35.1	41.1	41.6	39.2	37.3	34.3	27.6	21.3
		MCWB	9.6	11.3	12.2	14.2	14.9	17.4	18.9	18.4	17.3	15.7	12.7	10.1
	2%	DB	18.2	22.9	26.2	29.9	33.0	38.0	39.1	38.0	35.9	32.4	24.9	17.9
		MCWB	8.2	9.9	11.3	13.5	14.7	16.6	18.0	17.8	16.6	14.8	11.1	8.4
	5%	DB	17.0	21.1	23.9	27.8	31.4	36.9	37.8	37.3	34.1	30.6	22.5	16.3
		MCWB	7.6	9.2	10.7	12.2	14.4	16.4	17.6	17.3	16.2	13.9	10.2	7.3
	10%	DB	14.7	18.6	21.9	25.7	29.0	34.5	36.7	36.0	32.8	27.8	20.0	14.1
		MCWB	7.0	8.5	10.2	11.5	13.8	15.8	17.2	17.2	16.0	12.8	9.8	6.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.5	12.9	13.6	15.6	18.1	19.0	20.9	22.3	19.8	17.4	14.3	11.9
		MCDB	15.8	22.1	24.0	28.5	29.3	33.7	32.9	30.5	30.3	24.2	22.4	16.2
	2%	WB	10.7	11.5	12.4	14.0	16.5	18.1	20.0	20.5	18.9	16.1	12.9	10.7
		MCDB	15.8	18.1	23.0	27.4	27.8	34.4	32.3	31.2	30.3	26.5	20.6	15.2
	5%	WB	9.3	10.3	11.5	13.0	15.3	17.4	19.3	19.5	18.1	15.1	11.6	9.3
		MCDB	13.7	17.7	21.7	25.5	28.2	34.2	32.6	32.3	30.3	26.6	20.2	13.8
	10%	WB	7.9	9.4	10.6	12.1	14.2	16.5	18.6	18.7	17.2	14.2	10.4	7.6
		MCDB	12.9	16.9	19.9	23.6	27.8	32.3	33.3	32.6	30.0	26.0	18.4	12.8
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	13.4	14.1	15.1	15.5	17.0	16.5	16.5	15.9	15.5	14.4	12.4
		MCDBR	15.9	16.7	17.6	18.0	17.5	18.6	17.6	17.4	16.9	17.2	17.0	15.1
		MCWBR	8.7	7.8	7.6	7.3	6.4	6.4	6.2	5.8	5.9	7.1	8.4	8.3
	5% WB	MCDBR	12.2	13.7	15.6	16.2	16.0	17.0	14.7	14.4	14.7	14.5	14.3	12.1
		MCWBR	7.3	6.9	7.2	6.9	6.2	6.2	5.7	5.7	5.5	6.2	7.3	7.0
(40) Clear-Sky Solar Irradiance	taub		0.266	0.275	0.291	0.309	0.318	0.320	0.345	0.328	0.312	0.298	0.269	0.269
	taud		2.577	2.569	2.507	2.456	2.458	2.471	2.453	2.489	2.517	2.534	2.601	2.589
	Ebn at Noon		960	985	988	976	964	957	931	945	951	943	949	937
	Edh at Noon		76	86	100	111	112	111	112	106	98	88	74	72
(44) All-Sky Solar Radiation	RadAvg		3.23	4.18	5.66	6.95	7.79	8.41	7.79	7.33	6.31	4.84	3.63	2.94
	RadStd		0.23	0.39	0.26	0.25	0.30	0.33	0.36	0.24	0.17	0.24	0.14	0.19

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	+0.66	N/A	-1.81	+0.69	N/A	N/A	N/A	-118	+234	+119

Nomenclature: See separate page